STA2023 Exam 2 — every scenario, solved step by step

Study copy (too long to bring in; the two-sided cheat sheet is the one you bring). One box per way a question can be phrased. Each box is complete on its own: the words → what they really ask → numbered steps with the real math → the boxed answer → the sentence to write. Same numbers as the practice exam, so on the real exam you keep the steps and swap the numbers.

Question 1 — two-way table (or / and / given / not / two picks)

> Employment status of 78 residents.
> Males / Females — Full-time 22 / 18 (row total 40) · Part-time 6 / 10 (16) · Unemployed 3 / 2 (5) · Not in labor force 5 / 12 (17).
> Column totals: males 36, females 42. Grand total 78.

The recipe for every table question

1. Find the numbers: the cell, the row total, the column total, the grand total.
2. Pick the rule from the key word: **is A** → A's total ÷ 78 · **A or B** → add the two totals, subtract the overlap cell once · **A and B** → the one cell ÷ 78 · **A given B** → the cell ÷ B's total (not 78) · **not A** → 1 − P(A).
3. Write the fraction, then the decimal to 4 places, box it.

is male

Out of everybody, what fraction are male? One category → its total ÷ grand total.

Step 1 · Find the numbers Male is a column; its total is 36. Grand total 78.

Step 2 · Rule P(A) = (number in A) ÷ (total).

$$P(\text{male}) = \frac{\text{males}}{\text{total}}$$

Step 3 · Plug in and compute

$$P(\text{male}) = \frac{36}{78} = 0.4615$$

0.4615

Write on the exam: P(male) = 36/78 = 0.4615.

is part-time

A row: add both cells of the row, then ÷ 78.

Step 1 · Find the numbers Part-time row: 6 males + 10 females.

$$6 + 10 = 16$$

Step 2 · Rule

$$P(\text{part-time}) = \frac{\text{part-time total}}{\text{total}}$$

Step 3 · Plug in and compute

$$P(\text{part-time}) = \frac{16}{78} = 0.2051$$

0.2051

is NOT male

The complement: everything except male.

Step 1 · Find the numbers Males 36, total 78.

Step 2 · Rule P(not A) = 1 − P(A).

$$P(\text{not male}) = 1 - P(\text{male})$$

Step 3 · Plug in and compute

$$P(\text{not male}) = 1 - \frac{36}{78} = \frac{42}{78} = 0.5385$$

0.5385

Watch out: Same thing as counting the females directly: 42 ÷ 78.

male OR unemployed

Either group, or both. Adding the two totals counts the overlap (male AND unemployed) twice, so subtract it once.

Step 1 · Find the numbers Males 36. Unemployed 5. Overlap (unemployed males) 3.

Step 2 · Rule

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

Step 3 · Plug in

$$P(\text{M or U}) = \frac{36}{78} + \frac{5}{78} - \frac{3}{78}$$

Step 4 · Compute

$$= \frac{36 + 5 - 3}{78} = \frac{38}{78} = 0.4872$$

0.4872

Write on the exam: P(M or U) = 36/78 + 5/78 − 3/78 = 38/78 = 0.4872.

Watch out: Forgetting the overlap gives 41/78 = 0.5256 ×. Check: 36 males + the 2 unemployed women = 38 people ✓

full-time OR part-time (two rows of the same variable)

They can't both happen at once (nobody is in two rows), so the overlap is 0: just add.

Step 1 · Find the numbers Row totals 40 and 16. Overlap 0.

Step 2 · Rule

$$P(A \text{ or } B) = P(A) + P(B) - 0$$

Step 3 · Plug in and compute

$$P(\text{FT or PT}) = \frac{40}{78} + \frac{16}{78} = \frac{56}{78} = 0.7179$$

0.7179

female AND not in the labor force

Both at once → the single cell where the row and the column meet.

Step 1 · Find the numbers Female column, Not-in-labor-force row: 12.

Step 2 · Rule

$$P(A \text{ and } B) = \frac{\text{the one cell}}{\text{total}}$$

Step 3 · Plug in and compute

$$P(\text{F and NLF}) = \frac{12}{78} = 0.1538$$

0.1538

Watch out: "and" never adds anything — it is one cell.

part-time GIVEN female

You already know she is female, so the world shrinks to the 42 females. Divide by 42, not 78.

Step 1 · Find the numbers Part-time females: 10. All females: 42.

Step 2 · Rule

$$P(A \mid B) = \frac{\text{cell}}{\text{total of the given group } B}$$

Step 3 · Plug in and compute

$$P(\text{PT} \mid \text{F}) = \frac{10}{42} = 0.2381$$

0.2381

Write on the exam: P(PT | F) = 10/42 = 0.2381.

Watch out: 10/78 = 0.1282 is the classic error. The word after "given" tells you the denominator.

female GIVEN part-time

Other direction: now the given group is the 16 part-timers.

Step 1 · Find the numbers Part-time females: 10. All part-timers: 16.

Step 2 · Rule

$$P(\text{F} \mid \text{PT}) = \frac{\text{cell}}{\text{part-time total}}$$

Step 3 · Plug in and compute

$$P(\text{F} \mid \text{PT}) = \frac{10}{16} = 0.6250$$

0.6250

unemployed GIVEN male

Given group = the 36 males.

Step 1 · Find the numbers Unemployed males: 3. All males: 36.

Step 2 · Rule and compute

$$P(\text{U} \mid \text{M}) = \frac{3}{36} = 0.0833$$

0.0833

are "part-time" and "female" independent?

Independent = knowing one does not change the other. Test: is P(PT | F) the same number as P(PT)?

Step 1 · Compute the conditional

$$P(\text{PT} \mid \text{F}) = \frac{10}{42} = 0.2381$$

Step 2 · Compute the plain probability

$$P(\text{PT}) = \frac{16}{78} = 0.2051$$

Step 3 · Compare

$$0.2381 \neq 0.2051 \;\Rightarrow\; \text{dependent}$$

dependent (not independent)

Write on the exam: P(PT | F) = 0.2381 ≠ P(PT) = 0.2051, so the events are not independent.

Watch out: Equal → independent. Either direction (P(F | PT) vs P(F)) gives the same verdict.

two residents picked WITHOUT replacement: both male

"Both" = AND for two picks → multiply. The second pick has one fewer male and one fewer person.

Step 1 · First pick

$$P(\text{1st male}) = \frac{36}{78}$$

Step 2 · Second pick, given the first was male 35 males left out of 77 people.

$$P(\text{2nd male} \mid \text{1st male}) = \frac{35}{77}$$

Step 3 · Multiply

$$P(\text{both male}) = \frac{36}{78} \cdot \frac{35}{77} = 0.2098$$

0.2098

two residents picked WITH replacement (or a tiny sample from a huge population): both male

Independent picks: the same probability twice.

Step 1 · Rule

$$P(\text{both}) = P(\text{male}) \cdot P(\text{male})$$

Step 2 · Plug in and compute

$$P(\text{both male}) = \frac{36}{78} \cdot \frac{36}{78} = 0.2130$$

0.2130

two residents, at least one male

"At least one" = 1 − P(none). None male = both female.

Step 1 · P(none) = both female (without replacement)

$$P(\text{both F}) = \frac{42}{78} \cdot \frac{41}{77} = 0.2867$$

Step 2 · Subtract from 1

$$P(\text{at least one male}) = 1 - 0.2867 = 0.7133$$

0.7133

Watch out: Keep full decimals until the end; rounding each fraction first gives 0.7132.

two residents, exactly one male

Male-then-female OR female-then-male: two orders, add them.

Step 1 · Order 1: M then F

$$\frac{36}{78} \cdot \frac{42}{77} = \frac{1512}{6006} = 0.2517$$

Step 2 · Order 2: F then M

$$\frac{42}{78} \cdot \frac{36}{77} = \frac{1512}{6006} = 0.2517$$

Step 3 · Add (keep the fractions so no rounding creeps in)

$$P(\text{exactly one male}) = \frac{1512}{6006} + \frac{1512}{6006} = \frac{3024}{6006} = 0.5035$$

0.5035

Watch out: Adding the rounded 0.2517 + 0.2517 gives 0.5034; the exact fractions give 0.5035. Either is accepted, but keep fractions when you can.

Question 2 — is it a valid probability distribution?

The recipe (write both checks every time)

1. Rule 1: every P(x) is between 0 and 1 (0 and 1 themselves are allowed).
2. Rule 2: add all the P(x); the sum must be exactly 1.
3. Verdict in a sentence: valid, or not valid and WHICH rule fails.

the probabilities add to less than 1 (the practice exam)

x: 0, 1, 2, 3, 4 with P(x): 0.34, 0.21, 0.13, 0.04, 0.01.

Step 1 · Rule 1 Every P(x) is between 0 and 1 ✓

Step 2 · Rule 2: add

$$0.34 + 0.21 + 0.13 + 0.04 + 0.01 = 0.73$$

Step 3 · Compare with 1

$$0.73 \neq 1 \;\Rightarrow\; \text{rule 2 fails}$$

NOT valid — the probabilities add to 0.73, not 1

Write on the exam: Not a valid distribution: ΣP(x) = 0.73 ≠ 1.

the probabilities add to more than 1

x: 1–5 with P(x): 0.15, 0.25, 0.30, 0.20, 0.12.

Step 1 · Rule 1 All between 0 and 1 ✓

Step 2 · Rule 2: add

$$0.15 + 0.25 + 0.30 + 0.20 + 0.12 = 1.02$$

Step 3 · Compare with 1

$$1.02 \neq 1 \;\Rightarrow\; \text{rule 2 fails}$$

NOT valid — they add to 1.02, not 1

one probability is negative (or above 1)

x: 1–4 with P(x): 0.5, 0.4, 0.2, −0.1 (notice they add to 1).

Step 1 · Rule 1 P(4) = −0.1 is below 0 × — rule 1 fails.

Step 2 · Rule 2: add

$$0.5 + 0.4 + 0.2 + (-0.1) = 1 \;\checkmark\; (\text{passes, but rule 1 already failed})$$

NOT valid — a probability can never be negative

Watch out: A sum of 1 does not rescue a negative entry (and an entry like 1.3 fails rule 1 the same way).

everything checks out

x: 1–4 with P(x): 0.2, 0.4, 0.3, 0.1.

Step 1 · Rule 1 All between 0 and 1 ✓

Step 2 · Rule 2: add

$$0.2 + 0.4 + 0.3 + 0.1 = 1 \;\checkmark$$

valid — each P(x) is between 0 and 1 and they add to 1

Write on the exam: Valid: 0 ≤ P(x) ≤ 1 for every x and ΣP(x) = 1.

find the missing probability

x: 0–4 with P(x): 0.34, 0.21, 0.13, 0.04, ? — the blank must make the total exactly 1.

Step 1 · Add the known ones

$$0.34 + 0.21 + 0.13 + 0.04 = 0.72$$

Step 2 · Subtract from 1

$$P(4) = 1 - 0.72 = 0.28$$

Step 3 · Check rule 1 0.28 is between 0 and 1 ✓

0.28

Question 3 — discrete probability distribution (Stanley Cup games)

x = number of games in the series: 4, 5, 6, 7 happened 20, 20, 26, 17 times (total 83).
P(x) = frequency ÷ 83: P(4) = 0.2410, P(5) = 0.2410, P(6) = 0.3133, P(7) = 0.2048. μ = 5.48, σ² = 1.1413, σ = 1.07.

The recipe for every probability here

1. Words → the list of x values. **at least / at most / "or more" / "or less"** keep the number IN; **more than / fewer than** leave it OUT.
2. Write P(x) = frequency ÷ 83 for each x on the list.
3. Add them (or, if the list is most of the table, 1 − the leftovers). 4 decimals, box it.

build the probability distribution (part a)

Each probability = how often it happened ÷ total.

Step 1 · Rule

$$P(x) = \frac{\text{frequency}}{\text{total}} = \frac{\text{frequency}}{83}$$

Step 2 · Each x

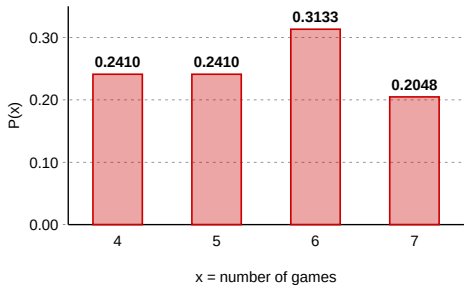
$$P(4) = \frac{20}{83} = 0.2410 \qquad P(5) = \frac{20}{83} = 0.2410$$

$$P(6) = \frac{26}{83} = 0.3133 \qquad P(7) = \frac{17}{83} = 0.2048$$

Step 3 · Check

$$0.2410 + 0.2410 + 0.3133 + 0.2048 = 1.0001 \approx 1 \text{ (rounding)}$$

the table of x and P(x) above



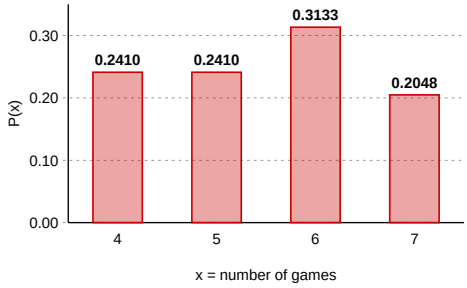
red bars = the ones you add

the graph (part b)

x across, P(x) up, one bar above each x with height P(x); label both axes. The tallest bar is the mode.

Step 1 · Draw Axis x: 4, 5, 6, 7. Axis P(x): 0 to 0.35. Bars of height 0.2410, 0.2410, 0.3133, 0.2048.

a bar for each x, labels on both axes



red bars = the ones you add

exactly 6

Just one value.

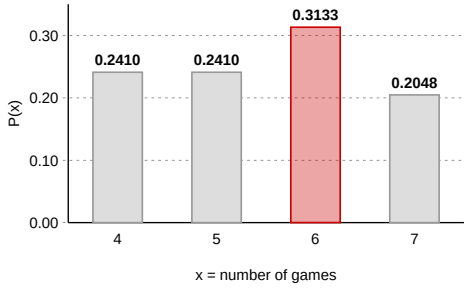
Step 1 · Words → list The x values that fit: **6**.

x = 4 5 **6** 7

Step 2 · Write each P(x) (frequency ÷ 83)

$$P(6) = \frac{26}{83} = 0.3133$$

0.3133



red bars = the ones you add

at least 6 (6 or more)

6 is IN → 6 and 7.

Step 1 · Words → list The x values that fit: **6, 7**.

x = 4 5 **6** **7**

Step 2 · Write each P(x) (frequency ÷ 83)

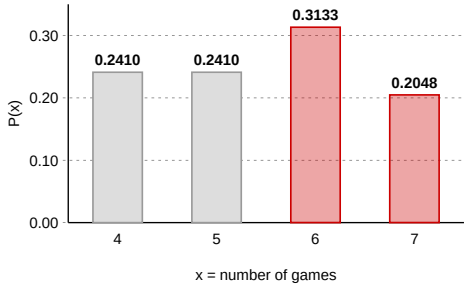
$$P(6) = \frac{26}{83} = 0.3133$$

$$P(7) = \frac{17}{83} = 0.2048$$

Step 3 · Add

$$P = 0.3133 + 0.2048 = 0.5181$$

0.5181



red bars = the ones you add

more than 6 (over 6)

6 is OUT → only 7.

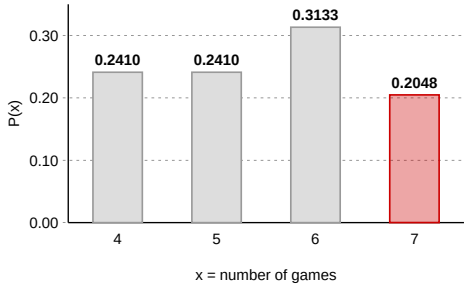
Step 1 · Words → list The x values that fit: **7**.

x = 4 5 6 **7**

Step 2 · Write each P(x) (frequency ÷ 83)

$$P(7) = \frac{17}{83} = 0.2048$$

0.2048



red bars = the ones you add

fewer than 6 (under 6, less than 6)

6 is OUT → 4 and 5.

Step 1 · Words → list The x values that fit: **4, 5**.

x = **4** **5** 6 7

Step 2 · Write each P(x) (frequency ÷ 83)

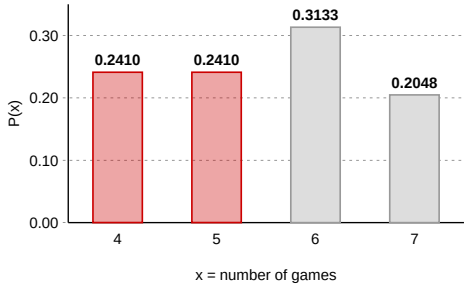
$$P(4) = \frac{20}{83} = 0.2410$$

$$P(5) = \frac{20}{83} = 0.2410$$

Step 3 · Add

$$P = 0.2410 + 0.2410 = 0.4820$$

0.4820



red bars = the ones you add

at most 5 (5 or fewer, no more than 5)

5 is IN → 4 and 5 (same list as "fewer than 6").

Step 1 · Words → list The x values that fit: **4, 5**.

x = **4** **5** 6 7

Step 2 · Write each P(x) (frequency ÷ 83)

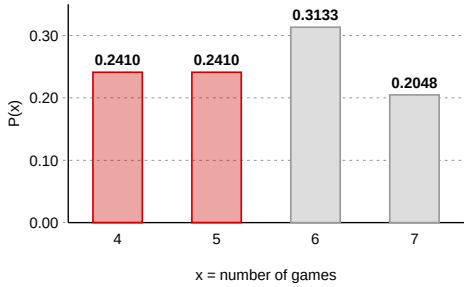
$$P(4) = \frac{20}{83} = 0.2410$$

$$P(5) = \frac{20}{83} = 0.2410$$

Step 3 · Add

$$P = 0.2410 + 0.2410 = 0.4820$$

0.4820



red bars = the ones you add

at most 6 (= fewer than 7 = not 7)

6 is IN → 4, 5, 6.

Step 1 · Words → list The x values that fit: **4, 5, 6**.

x = **4** **5** **6** 7

Step 2 · Write each P(x) (frequency ÷ 83)

$$P(4) = \frac{20}{83} = 0.2410$$
$$P(5) = \frac{20}{83} = 0.2410$$
$$P(6) = \frac{26}{83} = 0.3133$$

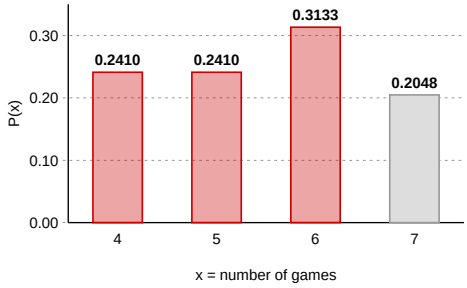
Step 3 · Add

$$P = 0.2410 + 0.2410 + 0.3133 = 0.7953$$

Step 4 · Shortcut (also accepted) The list is everything except 7, so you can also do 1 − P(7): the answers differ by 0.0001 only because each P(x) was rounded to 4 decimals — either one is accepted.

$$P = 1 - P(7) = 1 - 0.2048 = 0.7952$$

0.7953



red bars = the ones you add

at least 5 (= more than 4 = not 4)

5 is IN → 5, 6, 7.

Step 1 · Words → list The x values that fit: **5, 6, 7**.

x = 4 **5** **6** **7**

Step 2 · Write each P(x) (frequency ÷ 83)

$$P(5) = \frac{20}{83} = 0.2410$$
$$P(6) = \frac{26}{83} = 0.3133$$
$$P(7) = \frac{17}{83} = 0.2048$$

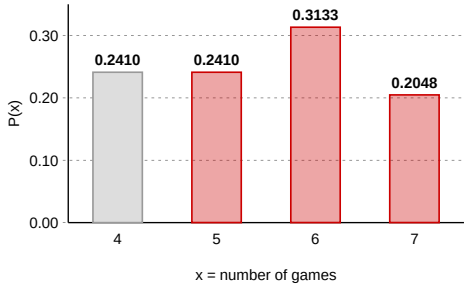
Step 3 · Add

$$P = 0.2410 + 0.3133 + 0.2048 = 0.7591$$

Step 4 · Shortcut (also accepted) The list is everything except 4, so you can also do 1 − P(4): the answers differ by 0.0001 only because each P(x) was rounded to 4 decimals — either one is accepted.

$$P = 1 - P(4) = 1 - 0.2410 = 0.7590$$

0.7591



red bars = the ones you add

between 5 and 6 inclusive

Both ends count → 5 and 6.

Step 1 · Words → list The x values that fit: **5, 6**.

x = 4 **5** **6** 7

Step 2 · Write each P(x) (frequency ÷ 83)

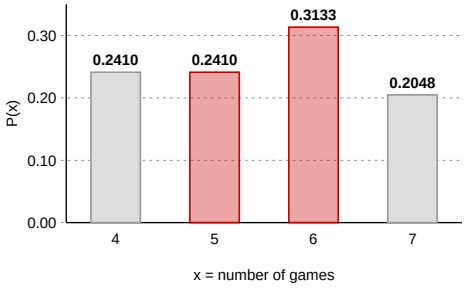
$$P(5) = \frac{20}{83} = 0.2410$$

$$P(6) = \frac{26}{83} = 0.3133$$

Step 3 · Add

$$P = 0.2410 + 0.3133 = 0.5543$$

0.5543



red bars = the ones you add

between 4 and 7 NOT inclusive (strictly between)

Ends are out → 5 and 6.

Step 1 · Words → list The x values that fit: **5, 6**.

x = 4 **5** **6** 7

Step 2 · Write each P(x) (frequency ÷ 83)

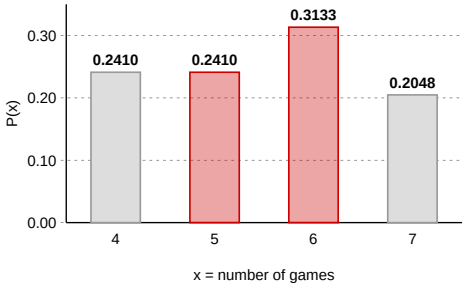
$$P(5) = \frac{20}{83} = 0.2410$$

$$P(6) = \frac{26}{83} = 0.3133$$

Step 3 · Add

$$P = 0.2410 + 0.3133 = 0.5543$$

0.5543



red bars = the ones you add

5 or 7

Two separate values → add them.

Step 1 · Words → list The x values that fit: **5, 7**.

x = 4 **5** 6 **7**

Step 2 · Write each P(x) (frequency ÷ 83)

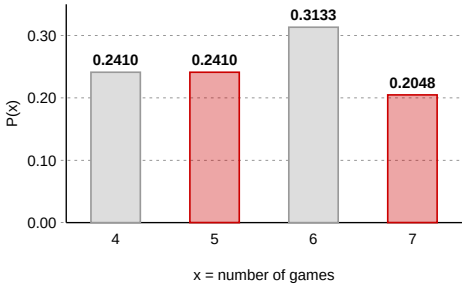
$$P(5) = \frac{20}{83} = 0.2410$$

$$P(7) = \frac{17}{83} = 0.2048$$

Step 3 · Add

$$P = 0.2410 + 0.2048 = 0.4458$$

0.4458



red bars = the ones you add

mean / expected value E(X) (part c)

Weighted average: each x times its probability, then add.

Step 1 · Rule

$$\mu = E(X) = \sum x \cdot P(x)$$

Step 2 · Plug in

$$\mu = 4(0.2410) + 5(0.2410) + 6(0.3133) + 7(0.2048)$$

Step 3 · Multiply each

$$= 0.9640 + 1.2050 + 1.8798 + 1.4336$$

Step 4 · Add

$$\mu = 5.48$$

μ = E(X) = 5.48 games

Write on the exam: Over many series, the average length is about 5.48 games.

variance and standard deviation (part d)

Subtract the mean, square, multiply by P(x), add (= variance), take √ (= SD).

Step 1 · Rule

$$\sigma^2 = \sum (x - \mu)^2 \cdot P(x) \qquad \sigma = \sqrt{\sigma^2}$$

Step 2 · Each term (μ = 5.48)

$$(4 - 5.48)^2(0.2410) = (-1.48)^2(0.2410) = 2.1904(0.2410) = 0.5279$$
$$(5 - 5.48)^2(0.2410) = (-0.48)^2(0.2410) = 0.2304(0.2410) = 0.0555$$
$$(6 - 5.48)^2(0.3133) = (0.52)^2(0.3133) = 0.2704(0.3133) = 0.0847$$
$$(7 - 5.48)^2(0.2048) = (1.52)^2(0.2048) = 2.3104(0.2048) = 0.4732$$

Step 3 · Add → variance

$$\sigma^2 = 0.5279 + 0.0555 + 0.0847 + 0.4732 = 1.1413$$

Step 4 · Square root → SD

$$\sigma = \sqrt{1.1413} = 1.07$$

σ² = 1.1413, σ = 1.07 games

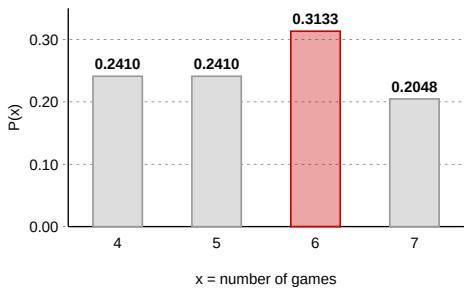
Watch out: If the question says variance, stop before the √. No ÷ 4 and no ÷ 83 here — the P(x) do the averaging.

most likely number of games (the mode)

The x with the biggest P(x).

Step 1 · Compare the P(x) 0.2410, 0.2410, 0.3133, 0.2048 → the biggest is 0.3133 at x = 6.

6 games



red bars = the ones you add

is a 7-game series unusual?

Two accepted tests: probability below 0.05, or x outside μ ± 2σ.

Step 1 · Probability test

$$P(7) = 0.2048 \text{ is not below } 0.05$$

Step 2 · Range test

$$\mu \pm 2\sigma = 5.48 \pm 2(1.07) = 3.34 \text{ to } 7.62$$

Step 3 · Decide 7 is inside 3.34 to 7.62 and 0.2048 ≥ 0.05.

not unusual

Write on the exam: P(7) = 0.2048 > 0.05, so a 7-game series is not unusual.

Question 4 — binomial (60% of women say yes; a sample of 10)

n = 10 trials (the 10 women), p = 0.60 (a "yes"), 1 − p = 0.40. X = how many of the 10 say yes. Formula: $P(x) = {}_nC_x \cdot p^x \cdot (1-p)^{n-x}$.

All eleven values, for checking: P(0) = 0.0001, P(1) = 0.0016, P(2) = 0.0106, P(3) = 0.0425, P(4) = 0.1115, P(5) = 0.2007, P(6) = 0.2508, P(7) = 0.2150, P(8) = 0.1209, P(9) = 0.0403, P(10) = 0.0060.

The recipe for every binomial probability (do these 5 steps every time)

1. **Words → list of x.** at most / at least / "or fewer" / "or more" keep the number IN; fewer than / more than leave it OUT; exactly = one value; at least one = 1 − P(0).
2. **Short list (1–4 values) → compute each and add. Long list → compute the OTHER side and do 1 − (that).**
3. **Each P(x) with the formula:** ${}_nC_x = n! \div (x!(n-x)!) \rightarrow p^x \rightarrow (1-p)^{n-x} \rightarrow$ multiply the three.
4. **Add** the terms; if you used the other side, **subtract from 1**.
5. 4 decimals, box it, write the sentence.

exactly 3

One value.

Step 1 · Words → list One value. The x values that fit: **3**.

`x = 0 1 2 3 4 5 6 7 8 9 10`

Step 2 · Short or long? 1 value — just compute it with the formula.

Step 3 · Set up

$$n = 10, \quad p = 0.6, \quad 1 - p = 0.4, \qquad P(x) = {}_nC_x\, p^x (1-p)^{n-x}$$

Step 4 · Compute P(3)

$$P(3) = {}_{10}C_3\, (0.6)^3\, (0.4)^7$$
$${}_{10}C_3 = \frac{10!}{3!\,7!} = \frac{10 \cdot 9 \cdot 8}{3 \cdot 2 \cdot 1} = 120$$
$$(0.6)^3 = 0.216 \qquad (0.4)^7 = 0.0016384$$
$$P(3) = 120 \times 0.216 \times 0.0016384 = 0.0425$$

0.0425

Write on the exam: P(3) = $_{10}C_3(0.6)^3(0.4)^7$ = 0.0425.

Watch out: The power on 0.4 is n − x = 7, not 3. Calculator check: nCr(10,3) × 0.6^3 × 0.4^7.

fewer than 3 (under 3, less than 3)

3 is OUT.

Step 1 · Words → list 3 is OUT. The x values that fit: **0, 1, 2**.

`x = 0 1 2 3 4 5 6 7 8 9 10`

Step 2 · Short or long? 3 values — short list: compute each one and add.

Step 3 · Set up

$$n = 10, \quad p = 0.6, \quad 1 - p = 0.4, \qquad P(x) = {}_nC_x\, p^x (1-p)^{n-x}$$

Step 4 · Compute P(0)

$$P(0) = {}_{10}C_0\, (0.6)^0\, (0.4)^{10}$$
$${}_{10}C_0 = \frac{10!}{0!\,10!} = 1$$
$$(0.6)^0 = 1 \qquad (0.4)^{10} = 0.0001048576$$
$$P(0) = 1 \times 1 \times 0.0001048576 = 0.0001$$

Step 5 · Compute P(1)

$$P(1) = {}_{10}C_1\, (0.6)^1\, (0.4)^9$$
$${}_{10}C_1 = \frac{10!}{1!\,9!} = \frac{10}{1} = 10$$
$$(0.6)^1 = 0.6 \qquad (0.4)^9 = 0.000262144$$
$$P(1) = 10 \times 0.6 \times 0.000262144 = 0.0016$$

Step 6 · Compute P(2)

$$P(2) = {}_{10}C_2\,(0.6)^2\,(0.4)^8$$
$${}_{10}C_2 = \frac{10!}{2!\,8!} = \frac{10\cdot 9}{2\cdot 1} = 45$$
$$(0.6)^2 = 0.36 \qquad (0.4)^8 = 0.00065536$$
$$P(2) = 45 \times 0.36 \times 0.00065536 = 0.0106$$

Step 7 · Add

$$P(X<3) = 0.0001 + 0.0016 + 0.0106 = 0.0123$$

0.0123

Write on the exam: P(X < 3) = P(0) + P(1) + P(2) = 0.0123.

at most 3 (3 or fewer, no more than 3, up to 3)

3 is IN.

Step 1 · Words → list 3 is IN. The x values that fit: **0, 1, 2, 3**.

x = **0 1 2 3** 4 5 6 7 8 9 10

Step 2 · Short or long? 4 values — short list: compute each one and add.

Step 3 · Set up

$$n = 10, \quad p = 0.6, \quad 1-p = 0.4, \qquad P(x) = {}_nC_x\,p^x(1-p)^{n-x}$$

Step 4 · Compute P(0)

$$P(0) = {}_{10}C_0\,(0.6)^0\,(0.4)^{10}$$
$${}_{10}C_0 = \frac{10!}{0!\,10!} = 1$$
$$(0.6)^0 = 1 \qquad (0.4)^{10} = 0.0001048576$$
$$P(0) = 1 \times 1 \times 0.0001048576 = 0.0001$$

Step 5 · Compute P(1)

$$P(1) = {}_{10}C_1\,(0.6)^1\,(0.4)^9$$
$${}_{10}C_1 = \frac{10!}{1!\,9!} = \frac{10}{1} = 10$$
$$(0.6)^1 = 0.6 \qquad (0.4)^9 = 0.000262144$$
$$P(1) = 10 \times 0.6 \times 0.000262144 = 0.0016$$

Step 6 · Compute P(2)

$$P(2) = {}_{10}C_2\,(0.6)^2\,(0.4)^8$$
$${}_{10}C_2 = \frac{10!}{2!\,8!} = \frac{10\cdot 9}{2\cdot 1} = 45$$
$$(0.6)^2 = 0.36 \qquad (0.4)^8 = 0.00065536$$
$$P(2) = 45 \times 0.36 \times 0.00065536 = 0.0106$$

Step 7 · Compute P(3)

$$P(3) = {}_{10}C_3\,(0.6)^3\,(0.4)^7$$
$${}_{10}C_3 = \frac{10!}{3!\,7!} = \frac{10\cdot 9\cdot 8}{3\cdot 2\cdot 1} = 120$$
$$(0.6)^3 = 0.216 \qquad (0.4)^7 = 0.0016384$$
$$P(3) = 120 \times 0.216 \times 0.0016384 = 0.0425$$

Step 8 · Add

$$P(X \le 3) = 0.0001 + 0.0016 + 0.0106 + 0.0425 = 0.0548$$

0.0548

Write on the exam: P(X ≤ 3) = P(0) + P(1) + P(2) + P(3) = 0.0548.

more than 3 (over 3)

3 is OUT, so the list is 4 through 10.

Step 1 · Words → list 3 is OUT, so the list is 4 through 10. The x values that fit: **4, 5, 6, 7, 8, 9, 10**.

`x = 0 1 2 3 4 5 6 7 8 9 10`

Step 2 · Short or long? 7 values is a long list. Compute the other side (x = 0, 1, 2, 3) and subtract from 1.

Step 3 · Set up

$$n = 10, \quad p = 0.6, \quad 1 - p = 0.4, \qquad P(x) = {}_nC_x\, p^x (1-p)^{n-x}$$

Step 4 · Compute P(0)

$$P(0) = {}_{10}C_0\, (0.6)^0\, (0.4)^{10}$$
$${}_{10}C_0 = \frac{10!}{0!\,10!} = 1$$
$$(0.6)^0 = 1 \qquad (0.4)^{10} = 0.0001048576$$
$$P(0) = 1 \times 1 \times 0.0001048576 = 0.0001$$

Step 5 · Compute P(1)

$$P(1) = {}_{10}C_1\, (0.6)^1\, (0.4)^9$$
$${}_{10}C_1 = \frac{10!}{1!\,9!} = \frac{10}{1} = 10$$
$$(0.6)^1 = 0.6 \qquad (0.4)^9 = 0.000262144$$
$$P(1) = 10 \times 0.6 \times 0.000262144 = 0.0016$$

Step 6 · Compute P(2)

$$P(2) = {}_{10}C_2\, (0.6)^2\, (0.4)^8$$
$${}_{10}C_2 = \frac{10!}{2!\,8!} = \frac{10 \cdot 9}{2 \cdot 1} = 45$$
$$(0.6)^2 = 0.36 \qquad (0.4)^8 = 0.00065536$$
$$P(2) = 45 \times 0.36 \times 0.00065536 = 0.0106$$

Step 7 · Compute P(3)

$$P(3) = {}_{10}C_3\, (0.6)^3\, (0.4)^7$$
$${}_{10}C_3 = \frac{10!}{3!\,7!} = \frac{10 \cdot 9 \cdot 8}{3 \cdot 2 \cdot 1} = 120$$
$$(0.6)^3 = 0.216 \qquad (0.4)^7 = 0.0016384$$
$$P(3) = 120 \times 0.216 \times 0.0016384 = 0.0425$$

Step 8 · Add

$$0.0001 + 0.0016 + 0.0106 + 0.0425 = 0.0548$$

Step 9 · Subtract from 1

$$P(X > 3) = 1 - 0.0548 = 0.9452$$

0.9452

Write on the exam: P(X > 3) = 1 − P(X ≤ 3) = 1 − 0.0548 = 0.9452.

Watch out: 1 − 0.0123 would be "at least 3". Decide IN or OUT first.

at least 3 (3 or more, no fewer than 3)

3 is IN, so the list is 3 through 10.

Step 1 · Words → list 3 is IN, so the list is 3 through 10. The x values that fit: **3, 4, 5, 6, 7, 8, 9, 10**.

`x = 0 1 2` **`3 4 5 6 7 8 9 10`**

Step 2 · Short or long? 8 values is a long list. Compute the other side (x = 0, 1, 2) and subtract from 1.

Step 3 · Set up

$$n = 10, \quad p = 0.6, \quad 1 - p = 0.4, \qquad P(x) = {}_nC_x\, p^x (1-p)^{n-x}$$

Step 4 · Compute P(0)

$$P(0) = {}_{10}C_0\, (0.6)^0\, (0.4)^{10}$$
$${}_{10}C_0 = \frac{10!}{0!\,10!} = 1$$
$$(0.6)^0 = 1 \qquad (0.4)^{10} = 0.0001048576$$
$$P(0) = 1 \times 1 \times 0.0001048576 = 0.0001$$

Step 5 · Compute P(1)

$$P(1) = {}_{10}C_1\, (0.6)^1\, (0.4)^9$$
$${}_{10}C_1 = \frac{10!}{1!\,9!} = \frac{10}{1} = 10$$
$$(0.6)^1 = 0.6 \qquad (0.4)^9 = 0.000262144$$
$$P(1) = 10 \times 0.6 \times 0.000262144 = 0.0016$$

Step 6 · Compute P(2)

$$P(2) = {}_{10}C_2\, (0.6)^2\, (0.4)^8$$
$${}_{10}C_2 = \frac{10!}{2!\,8!} = \frac{10 \cdot 9}{2 \cdot 1} = 45$$
$$(0.6)^2 = 0.36 \qquad (0.4)^8 = 0.00065536$$
$$P(2) = 45 \times 0.36 \times 0.00065536 = 0.0106$$

Step 7 · Add

$$0.0001 + 0.0016 + 0.0106 = 0.0123$$

Step 8 · Subtract from 1

$$P(X \geq 3) = 1 - 0.0123 = 0.9877$$

0.9877

Write on the exam: P(X ≥ 3) = 1 − P(X ≤ 2) = 1 − 0.0123 = 0.9877.

Watch out: The other side of "at least 3" is 0, 1, 2 — it does NOT include 3.

between 1 and 3 inclusive (from 1 to 3)

Both ends count.

Step 1 · Words → list Both ends count. The x values that fit: **1, 2, 3**.

`x = 0` **`1 2 3`** `4 5 6 7 8 9 10`

Step 2 · Short or long? 3 values — short list: compute each one and add.

Step 3 · Set up

$$n = 10, \quad p = 0.6, \quad 1 - p = 0.4, \qquad P(x) = {}_nC_x\, p^x (1-p)^{n-x}$$

Step 4 · Compute P(1)

$$P(1) = {}_{10}C_1\, (0.6)^1\, (0.4)^9$$
$${}_{10}C_1 = \frac{10!}{1!\,9!} = \frac{10}{1} = 10$$
$$(0.6)^1 = 0.6 \qquad (0.4)^9 = 0.000262144$$
$$P(1) = 10 \times 0.6 \times 0.000262144 = 0.0016$$

Step 5 · Compute P(2)

$$P(2) = {}_{10}C_2\, (0.6)^2\, (0.4)^8$$
$${}_{10}C_2 = \frac{10!}{2!\,8!} = \frac{10 \cdot 9}{2 \cdot 1} = 45$$
$$(0.6)^2 = 0.36 \qquad (0.4)^8 = 0.00065536$$
$$P(2) = 45 \times 0.36 \times 0.00065536 = 0.0106$$

Step 6 · Compute P(3)

$$P(3) = {}_{10}C_3\, (0.6)^3\, (0.4)^7$$
$${}_{10}C_3 = \frac{10!}{3!\,7!} = \frac{10 \cdot 9 \cdot 8}{3 \cdot 2 \cdot 1} = 120$$
$$(0.6)^3 = 0.216 \qquad (0.4)^7 = 0.0016384$$
$$P(3) = 120 \times 0.216 \times 0.0016384 = 0.0425$$

Step 7 · Add

$$P(1 \le X \le 3) = 0.0016 + 0.0106 + 0.0425 = 0.0547$$

0.0547

between 1 and 3 NOT inclusive (strictly between)

Ends are out.

Step 1 · Words → list Ends are out. The x values that fit: **2**.

`x = 0 1 2 3 4 5 6 7 8 9 10`

Step 2 · Short or long? 1 value — just compute it with the formula.

Step 3 · Set up

$$n = 10, \quad p = 0.6, \quad 1 - p = 0.4, \qquad P(x) = {}_nC_x\, p^x (1-p)^{n-x}$$

Step 4 · Compute P(2)

$$P(2) = {}_{10}C_2\, (0.6)^2\, (0.4)^8$$
$${}_{10}C_2 = \frac{10!}{2!\,8!} = \frac{10 \cdot 9}{2 \cdot 1} = 45$$
$$(0.6)^2 = 0.36 \qquad (0.4)^8 = 0.00065536$$
$$P(2) = 45 \times 0.36 \times 0.00065536 = 0.0106$$

0.0106

at least 1 (at least one says yes)

1 is IN, so the list is 1 through 10 — the other side is just x = 0.

Step 1 · Words → list 1 is IN, so the list is 1 through 10 — the other side is just x = 0. The x values that fit: **1, 2, 3, 4, 5, 6, 7, 8, 9, 10**.

`x = 0` **`1 2 3 4 5 6 7 8 9 10`**

Step 2 · Short or long? 10 values is a long list. Compute the other side (x = 0) and subtract from 1.

Step 3 · Set up

$$n = 10, \quad p = 0.6, \quad 1 - p = 0.4, \qquad P(x) = {}_nC_x\, p^x (1-p)^{n-x}$$

Step 4 · Compute P(0)

$$P(0) = {}_{10}C_0\, (0.6)^0\, (0.4)^{10}$$
$${}_{10}C_0 = \frac{10!}{0!\,10!} = 1$$
$$(0.6)^0 = 1 \qquad (0.4)^{10} = 0.0001048576$$
$$P(0) = 1 \times 1 \times 0.0001048576 = 0.0001$$

Step 5 · Subtract from 1

$$P(X \geq 1) = 1 - 0.0001 = 0.9999$$

0.9999

Write on the exam: P(X ≥ 1) = 1 − P(0) = 1 − 0.0001 = 0.9999.

none (nobody says yes, zero)

Just x = 0.

Step 1 · Words → list Just x = 0. The x values that fit: **0**.

`x =` **`0`** `1 2 3 4 5 6 7 8 9 10`

Step 2 · Short or long? 1 value — just compute it with the formula.

Step 3 · Set up

$$n = 10, \quad p = 0.6, \quad 1 - p = 0.4, \qquad P(x) = {}_nC_x\, p^x (1-p)^{n-x}$$

Step 4 · Compute P(0)

$$P(0) = {}_{10}C_0\, (0.6)^0\, (0.4)^{10}$$
$${}_{10}C_0 = \frac{10!}{0!\,10!} = 1$$
$$(0.6)^0 = 1 \qquad (0.4)^{10} = 0.0001048576$$
$$P(0) = 1 \times 1 \times 0.0001048576 = 0.0001$$

0.0001

all 10 say yes

Just x = 10.

Step 1 · Words → list Just x = 10. The x values that fit: **10**.

`x = 0 1 2 3 4 5 6 7 8 9 10`

Step 2 · Short or long? 1 value — just compute it with the formula.

Step 3 · Set up

$$n = 10, \quad p = 0.6, \quad 1 - p = 0.4, \qquad P(x) = {}_nC_x\, p^x (1-p)^{n-x}$$

Step 4 · Compute P(10)

$$P(10) = {}_{10}C_{10}\,(0.6)^{10}\,(0.4)^0$$
$${}_{10}C_{10} = \frac{10!}{10!\,0!} = 1$$
$$(0.6)^{10} = 0.0060466176 \qquad (0.4)^0 = 1$$
$$P(10) = 1 \times 0.0060466176 \times 1 = 0.0060$$

0.0060

at most 8 (a long list on the low side)

8 is IN, so the list is 0 through 8 — the other side is 9 and 10.

Step 1 · Words → list 8 is IN, so the list is 0 through 8 — the other side is 9 and 10. The x values that fit: **0, 1, 2, 3, 4, 5, 6, 7, 8**.

`x = 0 1 2 3 4 5 6 7 8 9 10`

Step 2 · Short or long? 9 values is a long list. Compute the other side (x = 9, 10) and subtract from 1.

Step 3 · Set up

$$n = 10, \quad p = 0.6, \quad 1 - p = 0.4, \qquad P(x) = {}_nC_x\, p^x (1-p)^{n-x}$$

Step 4 · Compute P(9)

$$P(9) = {}_{10}C_9\,(0.6)^9\,(0.4)^1$$
$${}_{10}C_9 = \frac{10!}{9!\,1!} = \frac{10}{1} = 10$$
$$(0.6)^9 = 0.010077696 \qquad (0.4)^1 = 0.4$$
$$P(9) = 10 \times 0.010077696 \times 0.4 = 0.0403$$

Step 5 · Compute P(10)

$$P(10) = {}_{10}C_{10}\,(0.6)^{10}\,(0.4)^0$$
$${}_{10}C_{10} = \frac{10!}{10!\,0!} = 1$$
$$(0.6)^{10} = 0.0060466176 \qquad (0.4)^0 = 1$$
$$P(10) = 1 \times 0.0060466176 \times 1 = 0.0060$$

Step 6 · Add

$$0.0403 + 0.0060 = 0.0463$$

Step 7 · Subtract from 1

$$P(X \le 8) = 1 - 0.0463 = 0.9537$$

0.9537

more than 8

8 is OUT → 9 and 10.

Step 1 · Words → list 8 is OUT → 9 and 10. The x values that fit: **9, 10**.

`x = 0 1 2 3 4 5 6 7 8 9 10`

Step 2 · Short or long? 2 values — short list: compute each one and add.

Step 3 · Set up

$$n = 10, \quad p = 0.6, \quad 1 - p = 0.4, \qquad P(x) = {}_nC_x\, p^x (1-p)^{n-x}$$

Step 4 · Compute P(9)

$$P(9) = {}_{10}C_9\, (0.6)^9\, (0.4)^1$$

$${}_{10}C_9 = \frac{10!}{9!\,1!} = \frac{10}{1} = 10$$

$$(0.6)^9 = 0.010077696 \qquad (0.4)^1 = 0.4$$

$$P(9) = 10 \times 0.010077696 \times 0.4 = 0.0403$$

Step 5 · Compute P(10)

$$P(10) = {}_{10}C_{10}\, (0.6)^{10}\, (0.4)^0$$

$${}_{10}C_{10} = \frac{10!}{10!\,0!} = 1$$

$$(0.6)^{10} = 0.0060466176 \qquad (0.4)^0 = 1$$

$$P(10) = 1 \times 0.0060466176 \times 1 = 0.0060$$

Step 6 · Add

$$P(X > 8) = 0.0403 + 0.0060 = 0.0463$$

0.0463

expected number (mean) and standard deviation for the sample of 10

Binomial shortcuts — no table needed.

Step 1 · Rule

$$\mu = np \qquad \sigma = \sqrt{np(1-p)}$$

Step 2 · Plug in

$$\mu = 10(0.6) = 6 \qquad \sigma = \sqrt{10(0.6)(0.4)} = \sqrt{2.4}$$

Step 3 · Compute

$$\sigma = 1.55$$

μ = 6, σ = 1.55

Write on the exam: We expect 6 of the 10 to say yes, give or take about 1.55.

expected number and SD for a sample of 200

Same shortcuts with n = 200.

Step 1 · Rule

$$\mu = np \qquad \sigma = \sqrt{np(1-p)}$$

Step 2 · Plug in

$$\mu = 200(0.6) = 120 \qquad \sigma = \sqrt{200(0.6)(0.4)} = \sqrt{48}$$

Step 3 · Compute

$$\sigma = 6.93$$

μ = 120, σ = 6.93

Watch out: 48 is the variance — take the square root for σ.

would 130 of 200 be unusual?

Usual range = μ ± 2σ (allowed when np(1 − p) ≥ 10).

Step 1 · Check the condition

$$np(1-p) = 200(0.6)(0.4) = 48 \geq 10 \checkmark$$

Step 2 · Usual range

$$\mu \pm 2\sigma = 120 \pm 2(6.93) = 120 \pm 13.86 = 106.14 \text{ to } 133.86$$

Step 3 · Decide 130 is inside the range.

not unusual

Write on the exam: 130 is inside μ ± 2σ = 106.1 to 133.9, so it is not unusual.

would only 2 of the 10 saying yes be unusual?

Unusual = the probability of that or something more extreme is below 0.05. "2 or fewer" = x = 0, 1, 2.

Step 1 · Words → list 2 or fewer → 0, 1, 2.

x = **0** **1** **2** 3 4 5 6 7 8 9 10

Step 2 · Add their probabilities (each worked in the "fewer than 3" box)

$$P(X \leq 2) = 0.0001 + 0.0016 + 0.0106 = 0.0123$$

Step 3 · Compare with 0.05

$$0.0123 < 0.05$$

yes, unusual

Write on the exam: P(X ≤ 2) = 0.0123 < 0.05, so it would be unusual.

is this a binomial experiment? (why the formula is allowed)

Four conditions — write all four.

Step 1 · Fixed number of trials n = 10 women.

Step 2 · Each trial has two outcomes yes / no.

Step 3 · Same probability every trial p = 0.60 for each woman.

Step 4 · Independent trials one woman's answer does not change another's (random sample).

yes — all four conditions hold

Watch out: A "not binomial" example: drawing cards without replacement from a small deck (p changes each draw).

Question 5 — uniform distribution (temperature from 70°F to 100°F)

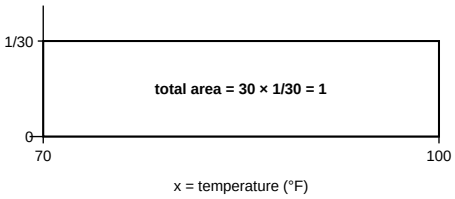
a = 70, b = 100, width b − a = 30. The graph is a flat rectangle of height 1/(b − a) = 1/30, so the total area is 30 × 1/30 = 1. Probability of any piece = (length of the piece) ÷ 30. No z-scores anywhere in this question.

The recipe for every uniform question

1. Write a, b and the width b − a.
2. Words → the piece: from c to d. "below 80" means 70 to 80; "above 90" means 90 to 100; anything outside 70–100 is cut off at 70 or 100.
3. P = (d − c) ÷ (b − a). 4 decimals, box it. (Backwards: x = a + area × (b − a).)

sketch / describe the pdf and give its height

A flat line from 70 to 100; zero everywhere else.



shaded area = the probability

Step 1 · Set up

$$a = 70, \quad b = 100, \quad b - a = 30$$

Step 2 · Height

$$\text{height} = \frac{1}{b-a} = \frac{1}{100-70} = \frac{1}{30} = 0.0333$$

Step 3 · Check the area

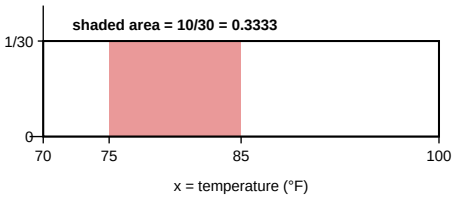
$$\text{area} = \text{width} \times \text{height} = 30 \times \frac{1}{30} = 1 \checkmark$$

rectangle from 70 to 100 with height 1/30

Write on the exam: f(x) = 1/30 for 70 ≤ x ≤ 100, and 0 otherwise.

between 75 and 85

A piece in the middle.



shaded area = the probability

Step 1 · Set up

$$a = 70, \quad b = 100, \quad b - a = 30$$

Step 2 · Words → piece From **75** to **85**; its length is 85 − 75 = 10.

Step 3 · Rule

$$P(c \le X \le d) = \frac{d - c}{b - a}$$

Step 4 · Plug in and compute

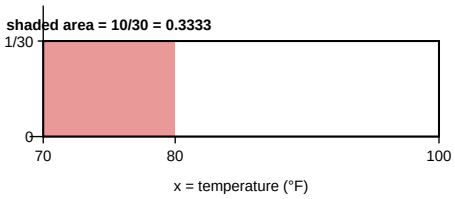
$$P(75 \le X \le 85) = \frac{85 - 75}{100 - 70} = \frac{10}{30} = 0.3333$$

0.3333

Write on the exam: P(75 ≤ X ≤ 85) = (85 − 75)/(100 − 70) = 10/30 = 0.3333.

below 80 (less than 80, at most 80, cooler than 80)

From the left end 70 up to 80.



shaded area = the probability

Step 1 · Set up

$$a = 70, \quad b = 100, \quad b - a = 30$$

Step 2 · Words → piece From **70** to **80**; its length is 80 − 70 = 10.

Step 3 · Rule

$$P(c \le X \le d) = \frac{d - c}{b - a}$$

Step 4 · Plug in and compute

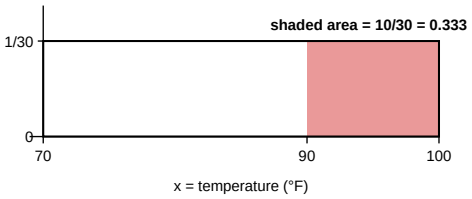
$$P(70 \le X \le 80) = \frac{80 - 70}{100 - 70} = \frac{10}{30} = 0.3333$$

0.3333

Watch out: P(X ≤ 80) is the identical number — one single point has no width, so P(X = 80) = 0.

above 90 (more than 90, at least 90, hotter than 90)

From 90 up to the right end 100.



shaded area = the probability

Step 1 · Set up

$$a = 70, \quad b = 100, \quad b - a = 30$$

Step 2 · Words → piece From **90** to **100**; its length is 100 − 90 = 10.

Step 3 · Rule

$$P(c \le X \le d) = \frac{d - c}{b - a}$$

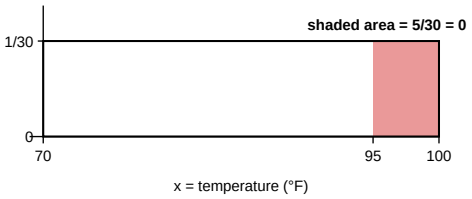
Step 4 · Plug in and compute

$$P(90 \le X \le 100) = \frac{100 - 90}{100 - 70} = \frac{10}{30} = 0.3333$$

0.3333

at least 95

From 95 to 100.



shaded area = the probability

Step 1 · Set up

$$a = 70, \quad b = 100, \quad b - a = 30$$

Step 2 · Words → piece From **95** to **100**; its length is 100 − 95 = 5.

Step 3 · Rule

$$P(c \le X \le d) = \frac{d - c}{b - a}$$

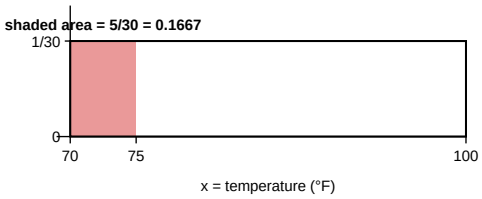
Step 4 · Plug in and compute

$$P(95 \le X \le 100) = \frac{100 - 95}{100 - 70} = \frac{5}{30} = 0.1667$$

0.1667

between 60 and 75 (60 is outside the range)

Only the part inside 70–100 counts, so the piece is 70 to 75.



shaded area = the probability

Step 1 · Set up

$$a = 70, \quad b = 100, \quad b - a = 30$$

Step 2 · Words → piece From **70** to **75**; its length is 75 − 70 = 5.

Step 3 · Rule

$$P(c \le X \le d) = \frac{d - c}{b - a}$$

Step 4 · Plug in and compute

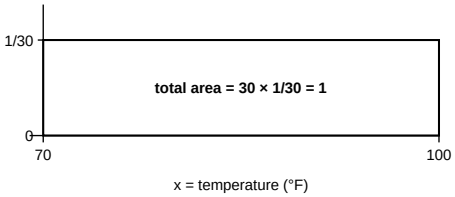
$$P(70 \le X \le 75) = \frac{75 - 70}{100 - 70} = \frac{5}{30} = 0.1667$$

0.1667

Watch out: Using 75 − 60 = 15 is wrong: nothing lives below 70.

exactly 80

A single point has no width, so no area.



shaded area = the probability

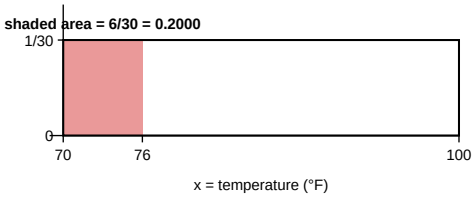
Step 1 · Rule

$$P(X = \text{one value}) = \frac{80 - 80}{30} = \frac{0}{30} = 0$$

0

20% of days are cooler than what temperature?

Backwards: the area to the LEFT is given (0.20); find the cut x.



shaded area = the probability

Step 1 · Set up

$$a = 70, \quad b = 100, \quad b - a = 30, \quad \text{area to the left} = 0.20$$

Step 2 · Rule (area = length ÷ 30, so length = area × 30)

$$x = a + (\text{area})(b - a)$$

Step 3 · Plug in and compute

$$x = 70 + 0.20(30) = 70 + 6 = 76$$

Step 4 · Check

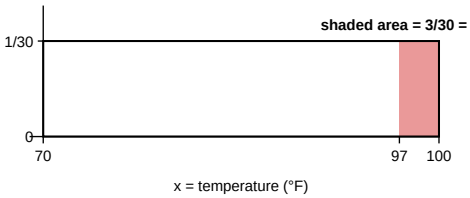
$$P(X < 76) = \frac{76 - 70}{30} = \frac{6}{30} = 0.20 \checkmark$$

76°F

Watch out: The 20% goes in as 0.20 — no z, no table, no hint.

the hottest 10% of days start at what temperature?

Top 10% means 90% are below it → area to the LEFT = 0.90.



shaded area = the probability

Step 1 · Set up

$$\text{area to the left} = 1 - 0.10 = 0.90$$

Step 2 · Rule

$$x = a + (\text{area})(b - a)$$

Step 3 · Plug in and compute

$$x = 70 + 0.90(30) = 70 + 27 = 97$$

Step 4 · Check

$$P(X > 97) = \frac{100 - 97}{30} = \frac{3}{30} = 0.10 \checkmark$$

97°F

the mean and standard deviation of the uniform (if asked)

Mean = the midpoint. SD = width ÷ √12.

Step 1 · Rule

$$\mu = \frac{a + b}{2} \qquad \sigma = \frac{b - a}{\sqrt{12}}$$

Step 2 · Plug in and compute

$$\mu = \frac{70 + 100}{2} = 85 \qquad \sigma = \frac{30}{\sqrt{12}} = 8.66$$

μ = 85, σ = 8.66

Question 6 — standard normal z (the hints give the area to the LEFT)

Hints as printed on the exam: left of −1.10 = 0.1357 · left of −0.60 = 0.2743 · left of 0.60 = 0.7257 · left of 1.45 = 0.9265 · (for the N(80, 10) part) left of −1.00 = 0.1587, left of 1.50 = 0.9332.

The recipe for every z question

1. Words → which area: LEFT of z (below, less than) · RIGHT of z (above, more than) · BETWEEN two z's · OUTSIDE two z's.
2. Read the hint(s). A hint is ALWAYS the area to the LEFT of that z.
3. LEFT → the hint itself · RIGHT → 1 − hint · BETWEEN → (left of the bigger z) − (left of the smaller z); bigger = further right, so −0.60 is bigger than −1.10 · OUTSIDE → left tail + right tail.
4. 4 decimals, box it. Backwards (area given, find z): make it a LEFT area, find the closest hint/table entry, read its z.

area to the LEFT of z = −1.10 (below −1.10)

Exactly what a hint gives.

Step 1 · Words → area "left of" → the hint is the answer.

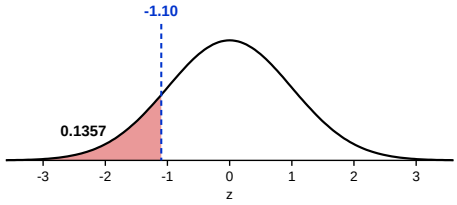
Step 2 · Read the hint

$$\text{left of } -1.10 = 0.1357$$

Step 3 · Rule and answer

$$P(Z < -1.10) = 0.1357$$

0.1357



shaded = the area asked for

Watch out: A hint for a negative z is already the left area — do not subtract it from 1.

area to the RIGHT of z = −1.10 (above −1.10)

Right = 1 − left.

Step 1 · Words → area "right of" → 1 − hint.

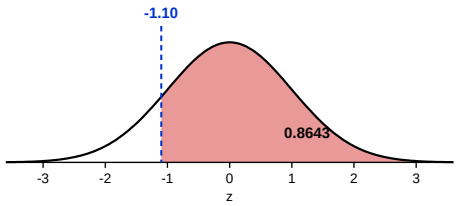
Step 2 · Read the hint

$$\text{left of } -1.10 = 0.1357$$

Step 3 · Rule and compute

$$P(Z > -1.10) = 1 - 0.1357 = 0.8643$$

0.8643



shaded = the area asked for

area to the RIGHT of z = 1.45

Right = 1 − left.

Step 1 · Words → area "right of" → 1 − hint.

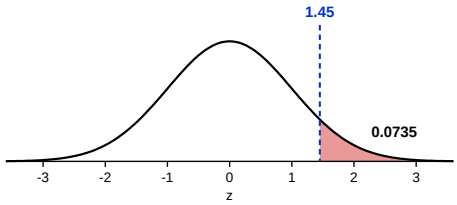
Step 2 · Read the hint

$$\text{left of } 1.45 = 0.9265$$

Step 3 · Rule and compute

$$P(Z > 1.45) = 1 - 0.9265 = 0.0735$$

0.0735



shaded = the area asked for

area BETWEEN z = −0.60 and z = 1.45

Bigger left area minus smaller left area.

Step 1 · Words → area "between" → (left of the bigger z) − (left of the smaller z). Bigger z = 1.45, smaller z = −0.60.

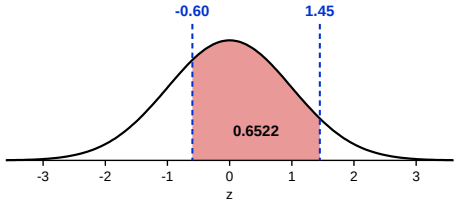
Step 2 · Read both hints

$$\text{left of } 1.45 = 0.9265 \qquad \text{left of } -0.60 = 0.2743$$

Step 3 · Rule and compute

$$P(-0.60 < Z < 1.45) = 0.9265 - 0.2743 = 0.6522$$

0.6522



shaded = the area asked for

Watch out: Subtract the areas, never the z-values.

between z = −1.10 and z = −0.60 (both negative)

Same rule; the bigger z is −0.60 (further right).

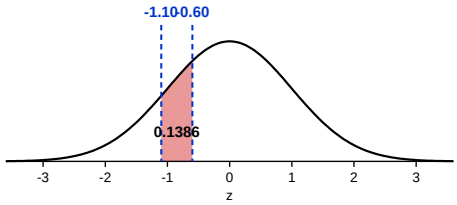
Step 1 · Read both hints

$$\text{left of } -0.60 = 0.2743 \qquad \text{left of } -1.10 = 0.1357$$

Step 2 · Rule and compute

$$P(-1.10 < Z < -0.60) = 0.2743 - 0.1357 = 0.1386$$

0.1386



shaded = the area asked for

between z = 0.60 and z = 1.45 (both positive)

Same rule.

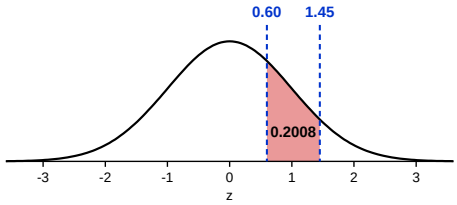
Step 1 · Read both hints

$$\text{left of } 1.45 = 0.9265 \qquad \text{left of } 0.60 = 0.7257$$

Step 2 · Rule and compute

$$P(0.60 < Z < 1.45) = 0.9265 - 0.7257 = 0.2008$$

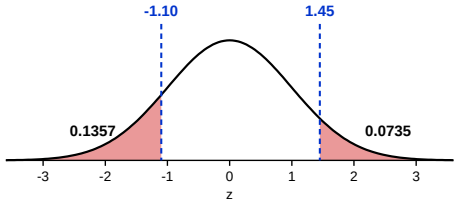
0.2008



shaded = the area asked for

OUTSIDE z = −1.10 and z = 1.45 (less than −1.10 OR more than 1.45; the two tails)

Left tail + right tail.



shaded = the area asked for

Step 1 · Left tail

$$P(Z < -1.10) = 0.1357$$

Step 2 · Right tail

$$P(Z > 1.45) = 1 - 0.9265 = 0.0735$$

Step 3 · Add

$$P(\text{outside}) = 0.1357 + 0.0735 = 0.2092$$

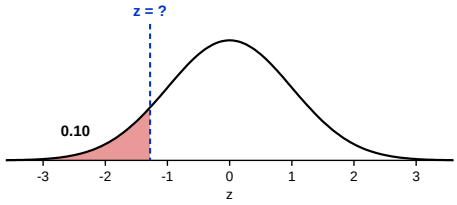
Step 4 · Same answer the other way

$$1 - P(\text{between}) = 1 - (0.9265 - 0.1357) = 1 - 0.7908 = 0.2092$$

0.2092

find the z with area 0.10 to its LEFT (the 10th percentile of z)

Backwards: the area is given; find the hint/table entry closest to 0.10 and read its z.



shaded = the area asked for

Step 1 · Words → left area

$$\text{area to the left} = 0.10$$

Step 2 · Find the closest entry 0.1003 is the closest (0.0985 is further).

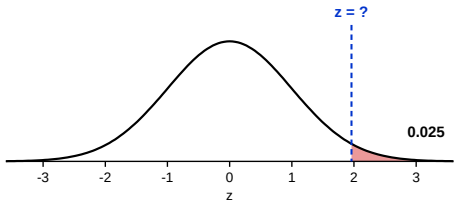
$$0.1003 \rightarrow z = -1.28$$

z = −1.28

Watch out: Left area below 0.5 → z is negative.

find the z with area 0.025 to its RIGHT

Convert to a LEFT area first.



shaded = the area asked for

Step 1 · Words → left area

$$\text{area to the left} = 1 - 0.025 = 0.975$$

Step 2 · Find the entry

$$0.9750 \rightarrow z = 1.96$$

z = 1.96

the two z's that cut off the middle 90%

0.05 left over in each tail.

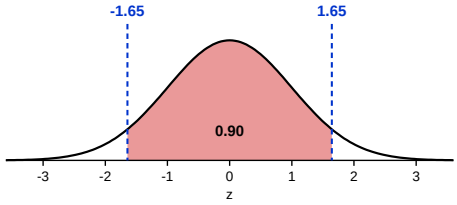
Step 1 · Words → left areas

$$\text{tails} = \frac{1 - 0.90}{2} = 0.05 \text{ each} \quad \Rightarrow \quad \text{left areas } 0.05 \text{ and } 0.95$$

Step 2 · Find the entries 0.95 sits exactly halfway between 0.9495 (z = 1.64) and 0.9505 (z = 1.65) → average 1.645; symmetry gives the other.

$$z = -1.645 \text{ and } z = 1.645$$

z = ±1.645



shaded = the area asked for

X ~ N(μ = 80, σ = 10): P(70 < X < 95)

Not standard: convert each x to z first, then BETWEEN.

Step 1 · Words → area "between" → bigger left − smaller left.

Step 2 · z for each end

$$z_1 = \frac{x - \mu}{\sigma} = \frac{70 - 80}{10} = -1.00 \qquad z_2 = \frac{95 - 80}{10} = 1.50$$

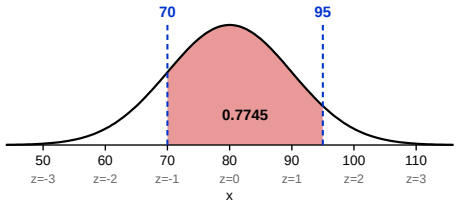
Step 3 · Read the hints

$$\text{left of } -1.00 = 0.1587 \qquad \text{left of } 1.50 = 0.9332$$

Step 4 · Rule and compute

$$P(70 < X < 95) = 0.9332 - 0.1587 = 0.7745$$

0.7745



x scale; z underneath

X ~ N(80, 10): P(X > 95)

Convert, then RIGHT = 1 − left.

Step 1 · z

$$z = \frac{95 - 80}{10} = 1.50$$

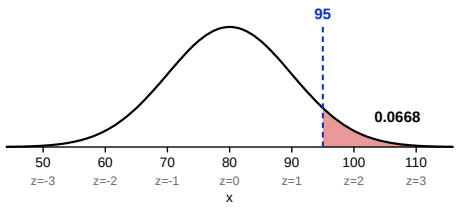
Step 2 · Read the hint

$$\text{left of } 1.50 = 0.9332$$

Step 3 · Rule and compute

$$P(X > 95) = 1 - 0.9332 = 0.0668$$

0.0668



x scale; z underneath

X ~ N(80, 10): P(X ≤ 70)

Convert, then LEFT = the hint.

Step 1 · z

$$z = \frac{70 - 80}{10} = -1.00$$

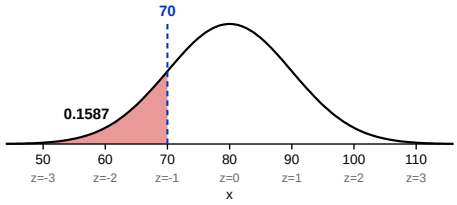
Step 2 · Read the hint

$$\text{left of } -1.00 = 0.1587$$

Step 3 · Rule and answer

$$P(X \le 70) = 0.1587$$

0.1587



x scale; z underneath

Question 7 — normal distribution in context (IQ scores: μ = 100, σ = 15)

Forward (x given, area wanted): z = (x − μ)/σ to 2 decimals → the hint gives the area to the LEFT → left / 1 − left / subtract.
Backward (percent given, x wanted): percent → area to the LEFT → the hint's z → x = μ + zσ.

The recipe for every normal word problem

1. Words → which area (below / above / between / outside), or backwards (a percent or percentile is given → find x).
2. Forward: z = (x − μ)/σ for each x, 2 decimals. Read the hint = area to the LEFT.
3. below → the hint · above → 1 − hint · between → bigger left − smaller left · outside → left tail + right tail. 4 decimals.
4. Backward: turn the percent into a LEFT area → the hint's z → x = μ + zσ (1 decimal).
5. Write the sentence in context ("…% of children score…").

above 125 (more than 125, at least 125, higher than 125)

Right tail.

Step 1 · Words → area "above" → 1 − (left of z).

Step 2 · z-score

$$z = \frac{x - \mu}{\sigma} = \frac{125 - 100}{15} = \frac{25}{15} = 1.67$$

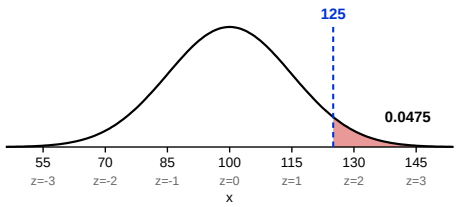
Step 3 · Read the hint

$$\text{left of } 1.67 = 0.9525$$

Step 4 · Rule and compute

$$P(X > 125) = 1 - 0.9525 = 0.0475$$

0.0475



IQ scale; z underneath

Write on the exam: About 4.75% of children have an IQ above 125.

Watch out: 0.9525 is the number-one wrong answer. Fewer than half score above 125, so the answer must be below 0.5.

below 90 (less than 90, at most 90, under 90)

Left area = the hint.

Step 1 · Words → area "below" → the hint itself.

Step 2 · z-score

$$z = \frac{90 - 100}{15} = \frac{-10}{15} = -0.67$$

Step 3 · Read the hint

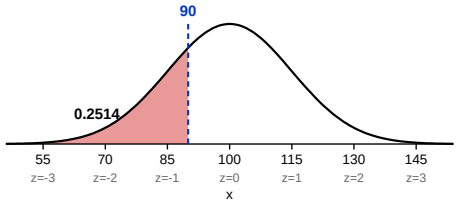
$$\text{left of } -0.67 = 0.2514$$

Step 4 · Rule and answer

$$P(X < 90) = 0.2514$$

0.2514

Write on the exam: About 25.14% of children have an IQ below 90.



IQ scale; z underneath

between 110 and 140

Two z's, two left areas, bigger minus smaller.

Step 1 · Words → area "between" → (left of bigger z) − (left of smaller z).

Step 2 · z for each end

$$z_1 = \frac{110 - 100}{15} = 0.67 \qquad z_2 = \frac{140 - 100}{15} = 2.67$$

Step 3 · Read the hints

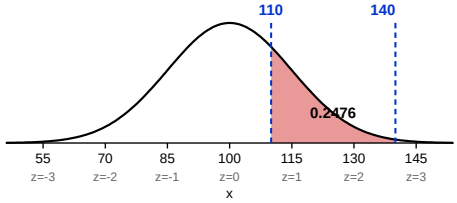
$$\text{left of } 0.67 = 0.7486 \qquad \text{left of } 2.67 = 0.9962$$

Step 4 · Rule and compute

$$P(110 < X < 140) = 0.9962 - 0.7486 = 0.2476$$

0.2476

Write on the exam: About 24.76% of children score between 110 and 140.



IQ scale; z underneath

outside 70 to 130 (below 70 OR above 130)

Two tails: left tail + right tail.

Step 1 · z for each end

$$z_1 = \frac{70 - 100}{15} = -2.00 \qquad z_2 = \frac{130 - 100}{15} = 2.00$$

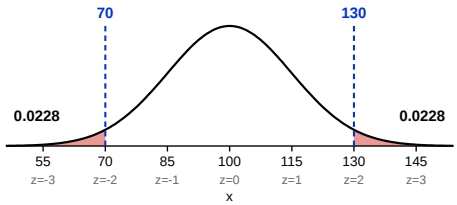
Step 2 · Read the hints

$$\text{left of } -2.00 = 0.0228 \qquad \text{left of } 2.00 = 0.9772$$

Step 3 · Left tail + right tail

$$P = 0.0228 + (1 - 0.9772) = 0.0228 + 0.0228 = 0.0456$$

0.0456



IQ scale; z underneath

exactly 100

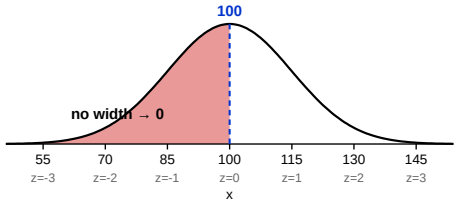
One single value has no area under a curve.

Step 1 · Rule

$$P(X = 100) = 0$$

0

Watch out: Any "exactly" with a continuous variable is 0; "at most 100" would be 0.5000.



IQ scale; z underneath

the 98th percentile (98% score below it)

Backwards: left area 0.98 → the hint gives z → x = μ + zσ.

Step 1 · Words → left area

$$\text{area to the left} = 0.98$$

Step 2 · Hint → z closest entry 0.9798

$$z = 2.05$$

Step 3 · Rule

$$x = \mu + z\sigma$$

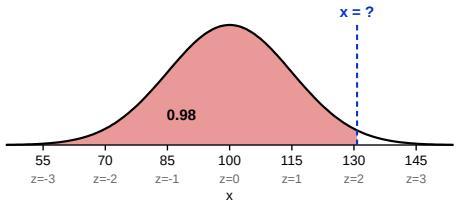
Step 4 · Plug in and compute

$$x = 100 + 2.05(15) = 100 + 30.75 = 130.75$$

130.75 ≈ 131

Write on the exam: An IQ of about 131 is the 98th percentile.

Watch out: Never put 0.98 into the z formula — it is an area, not a score.



IQ scale; z underneath

the top 5% cutoff (minimum score to be in the top 5%)

Top 5% → 95% below it → left area 0.95.

Step 1 · Words → left area

$$\text{area to the left} = 1 - 0.05 = 0.95$$

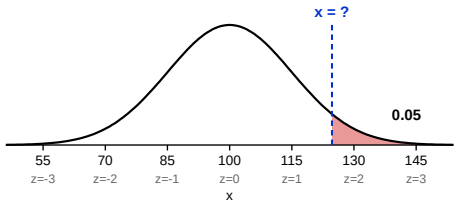
Step 2 · Hint → z 0.95 is halfway between 1.64 and 1.65

$$z = 1.645$$

Step 3 · Rule, plug in, compute

$$x = 100 + 1.645(15) = 100 + 24.675 = 124.675 \approx 124.7$$

124.7



IQ scale; z underneath

the bottom 10% cutoff (10th percentile)

Left area 0.10 → z is negative.

Step 1 · Words → left area

$$\text{area to the left} = 0.10$$

Step 2 · Hint → z closest entry 0.1003

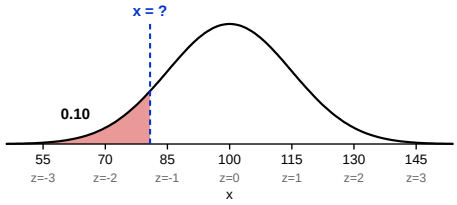
$$z = -1.28$$

Step 3 · Rule, plug in, compute

$$x = 100 + (-1.28)(15) = 100 - 19.2 = 80.8$$

80.8

Watch out: Keep the minus sign: the answer must be below the mean.



IQ scale; z underneath

the two scores holding the middle 95%

2.5% in each tail → left areas 0.025 and 0.975.

Step 1 · Words → left areas

$$\text{tails} = \frac{1 - 0.95}{2} = 0.025 \quad \Rightarrow \quad 0.025 \text{ and } 0.975$$

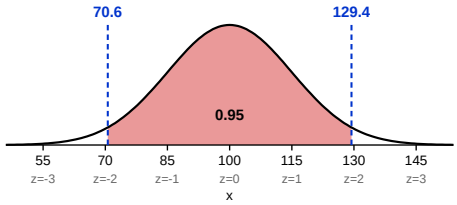
Step 2 · Hints → z

$$z = -1.96 \text{ and } z = 1.96$$

Step 3 · Rule, plug in, compute

$$x = 100 \pm 1.96(15) = 100 \pm 29.4 \quad \Rightarrow \quad 70.6 \text{ and } 129.4$$

70.6 to 129.4



IQ scale; z underneath

the score with 30% above it (70th percentile)

30% above → 70% below → left area 0.70.

Step 1 · Words → left area

$$\text{area to the left} = 1 - 0.30 = 0.70$$

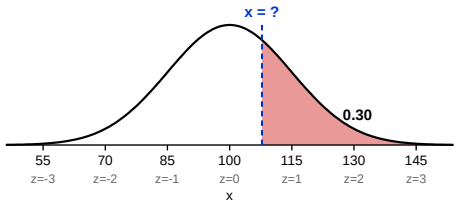
Step 2 · Hint → z closest entry 0.6985

$$z = 0.52$$

Step 3 · Rule, plug in, compute

$$x = 100 + 0.52(15) = 100 + 7.8 = 107.8$$

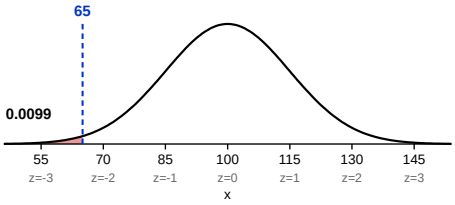
107.8



IQ scale; z underneath

is a score of 65 unusual?

Unusual = probability below 0.05 (or z beyond ±2).



IQ scale; z underneath

Step 1 · z-score

$$z = \frac{65 - 100}{15} = \frac{-35}{15} = -2.33$$

Step 2 · Read the hint

$$\text{left of } -2.33 = 0.0099$$

Step 3 · Compare

$$0.0099 < 0.05 \text{ (and } |-2.33| > 2)$$

yes, unusual

Write on the exam: P(X ≤ 65) = 0.0099 < 0.05, so a score of 65 is unusual.

interpret "the area above 125 is 0.0475" (two ways)

An area under the curve is both a proportion and a probability.

Step 1 · Proportion 4.75% of all children score above 125.

Step 2 · Probability A randomly selected child has a 0.0475 chance of scoring above 125.

write both sentences